

2024 F-350

MASTER REFERENCE MANUAL

VIN-specific ownership, towing, and service reference

Vehicle	Value
VIN	1FT8W3BNXRED73946
Model year	2024
Vehicle	Ford F-350 Super Duty
Assembly plant	Kentucky Truck Plant, Louisville, Kentucky
Engine	7.3L gasoline V8 (configuration record)
Bed / towing context	8-foot bed; factory fifth-wheel/gooseneck preparation and camera equipment recorded in project notes

Edition 1.0 - September 2026

This is a living reference, compiled from the established project conversation and the listed Ford documents. It summarizes rather than reproduces source manuals. Values marked “confirm with Ford WSM” are intentionally not service instructions until verified.

How to use this manual

- Use bookmarks or the linked table of contents to navigate by chapter.
- Treat “Ford OEM” as a source classification, not a substitute for consulting the original document for complete warnings and procedures.
- Record installed accessories, maintenance, scale weights, and module backups in the blank logs as work is performed.

Table of Contents

Table of Contents	2
1. Vehicle Information	4
Identity and configuration	4
Scope and confidence	4
Ownership record	4
2. Factory Build & Operation	5
Recorded factory equipment	5
Camera and trailer guidance quick reference	5
Driving systems notes	5
3. OEM Parts Catalog	6
Catalog status	6
Planned catalog fields	6
4. Maintenance & Service Log	7
Maintenance framework	7
Service log	7
5. Torque Specifications	8
Verified-status note	8
Provisional quick list - confirm with Ford WSM	8
Torque Atlas placeholder	8
6. Fluids & Capacities	9
Status	9
Planned table	9
7. Electrical Atlas	10
APIM / BCM rear-camera configuration	10
Electrical work placeholders	10
8. FORScan & Module Programming	11
Recovery-first workflow	11
Change log	11
9. Accessories & Modifications	12
Planned roadmap	12
Accessory installation record	12
10. Fifth-Wheel & Towing	13
OEM fifth-wheel hitch guidance	13

Hitching and backing checklist	13
Payload and loading	13
11. Troubleshooting	14
Placeholder symptom index	14
12. 3D-Printed Accessories	15
Design library	15
Design record template	15
13. Reference Documents	16
Source library	16
Future source intake	16
Appendix A. Future Workshop Manual Intake	17
Required additions	17
Revision record	17

1. Vehicle Information

Identity and configuration

Field	Recorded information
VIN	1FT8W3BNXRED73946
Manufacturer / type	Ford Motor Company / truck
Model year code	R = 2024
Plant code	E = Kentucky Truck Plant
Project configuration	F-350 Super Duty, 7.3L gasoline V8; XLT, 8-foot bed, fifth-wheel/gooseneck prep noted in prior project work
Dealer archive	Northside Ford Lincoln, Effingham, Illinois - original listing archive planned

Scope and confidence

This manual uses four labels: **Ford OEM** for official publications; **Recorded** for project-specific information; **Planned** for intended purchases or work; and **Confirm with Ford WSM** for data not yet validated against current service information.

Ownership record

Date	Mileage	Event / notes
2026-09-06		Master reference manual initial edition

2. Factory Build & Operation

Recorded factory equipment

- XLT trim and 7.3L gasoline configuration are recorded in the project conversation.
- Factory 360-degree camera package, bed camera, and fifth-wheel/gooseneck preparation are recorded as present.
- Factory in-bed 7-pin connector is recorded as present; it aligns with Ford's recommended driver-side bed location.

Camera and trailer guidance quick reference

- Available project-recorded views: rear view, 360-degree, rear split view, bed camera, picture-in-picture, and trailer guidance views when equipped.
- For fifth-wheel alignment: select the bed camera, use 360-degree view to monitor surroundings, confirm kingpin alignment, and then verify coupling before travel.
- Hitch-angle display: black line indicates actual trailer position; white line indicates predicted path. Yellow indicates approaching a difficult articulation angle; red indicates stop, pull forward, and realign.

Driving systems notes

- Adaptive cruise control offers selectable following gaps. The project notes caution against use when towing a trailer with an aftermarket electronic brake controller.
- Hill Descent Control operating range recorded from quick guide: 2-20 mph (3-32 km/h).
- Trail Control is low-speed off-road cruise control; braking reduces set speed without turning it off.

3. OEM Parts Catalog

Catalog status

Placeholder chapter. Add Ford and Motorcraft part numbers only after matching each item to the VIN/build and confirming current supersessions.

Planned catalog fields

System	Ford part no.	Motorcraft / equivalent	Interval / notes
Oil service			Confirm with WSM / owner manual
Air filtration			
Cabin air filter			
Brakes			
Trailer / hitch			

4. Maintenance & Service Log

Maintenance framework

- Use the official Owner’s Manual schedule as the primary source for normal and severe-duty intervals.
- Record date, mileage, parts used, fluid quantity, receipts, and any diagnostic results.
- Future additions: oil analysis, tire wear, alignment reports, battery testing, recalls, warranty work, and scale tickets.

Service log

Date	Mileage	Service / parts	Next due

5. Torque Specifications

Verified-status note

Important

The following values were previously recorded in the project conversation as known/common values, but must be checked against the current Ford Workshop Manual for this VIN before safety-critical use. Do not infer a specification for an unlisted fastener.

Provisional quick list - confirm with Ford WSM

Component	Previously recorded value	Status
Wheel lug nuts (18-inch steel/aluminum)	165 ft-lb (224 N-m)	Confirm with Ford WSM
7.3L engine oil drain plug	19 ft-lb (26 N-m)	Confirm with Ford WSM
7.3L spark plugs	15 ft-lb (20 N-m)	Confirm with Ford WSM
Ignition coil bolts	89 in-lb (10 N-m)	Confirm with Ford WSM
Battery terminal clamp nuts	44 in-lb (5 N-m)	Confirm with Ford WSM
Wiper arm nuts	30 ft-lb (40 N-m)	Confirm with Ford WSM
Receiver hitch mounting bolts	210 ft-lb (285 N-m)	Confirm with Ford WSM

Torque Atlas placeholder

- Future system sections: brakes, steering, front and rear suspension, drivetrain, 7.3L engine, cooling, body/frame, electrical, and towing equipment.
- Each final entry should include torque, units, fastener dimensions, sequence, angle, one-time-use status, thread treatment, and Ford WSM reference.

6. Fluids & Capacities

Status

Placeholder chapter. Populate exact capacities, approved fluids, and fill procedures from the Owner's Manual and Ford Workshop Manual for the truck's verified axle, transmission, and transfer-case configuration.

Planned table

System	Capacity	Specification	Source / revision
Engine oil			
Cooling system			
Transmission			
Transfer case			
Front / rear axle			
Brake fluid			

7. Electrical Atlas

APIM / BCM rear-camera configuration

- Reference REF-008 covers enabling or disabling an OEM rear camera kit through APIM and BCM configuration.
- Recorded post-programming verification: complete APIM programming; switch ignition off; open a front door and allow cluster/display power-down; switch ignition on; select Reverse with brake applied and parking brake engaged to verify image.
- This procedure is not currently needed for the truck's recorded factory camera equipment. It becomes relevant after APIM/BCM replacement, camera retrofit, a lost configuration, or relevant FORScan/FDRS programming.

Electrical work placeholders

- Fuse and relay directory; customer access wiring and upfitter circuits; grounds; connector pinouts; trailer wiring; camera wiring; CAN network notes.
- Before programming, save full module backups and record each change, date, tool version, and recovery steps.

8. FORScan & Module Programming

Recovery-first workflow

- Create and retain original As-Built/module backups before making changes.
- Log every change with before/after values, rationale, tool/interface version, and verification result.
- Use the camera configuration reference only for a documented camera/API/BCM situation; avoid changing unrelated configuration values.

Change log

Date	Module	Change	Backup location / verification

9. Accessories & Modifications

Planned roadmap

Phase	Planned items
Immediate	Sam's Club Plus membership (military discount); Michelin Defender LTX M/S2 tires; Covercraft Spacer Mesh PrecisionFit covers (front EM / rear AU); lace-on leather steering-wheel cover
Protection	Wheel-well liners; window deflectors; mud flaps; sill protection; LampGard; paint protection film; TowTector; tailgate lock; Console Vault
Towing	B&W; Companion OEM fifth-wheel hitch; 7-way extension/adapters; plug tester; chocks; levelers; infrared thermometer
Storage	Dee Zee under-seat drawer; bed cover (8-foot compatibility needs vendor confirmation); console and modular 3D-printed organizers
Maintenance / security	Grease tools, dielectric grease, cleaning supplies, spare adapters/pins; coupler/wheel locks and security cable

Accessory installation record

For each installed item, capture manufacturer, part number, vendor, date, cost, photos, torque/wiring instructions, installation notes, warranty, and removal process.

10. Fifth-Wheel & Towing

OEM fifth-wheel hitch guidance

- The Ford OEM fifth-wheel hitch reference supports use with an 8-foot pickup bed, matching the project-recorded truck configuration.
- Before working between truck and trailer: Park transmission, apply parking brake, chock both sides of trailer wheels, and support landing gear. Keep body parts out from between truck and trailer whenever practical.
- Maximum allowable trailer weight is controlled by the lowest-rated component among truck, hitch, trailer, and related equipment. Gross trailer weight includes cargo and consumables.

Hitching and backing checklist

Step	Action
1	Inspect hitch, pins, chains, connector, landing gear, tires, and surrounding area.
2	Use bed camera for kingpin alignment and 360-degree view for perimeter awareness.
3	Confirm coupling/lock engagement and perform a pull test as specified by hitch/trailer instructions.
4	Connect breakaway cable, safety equipment, and electrical connector; confirm lights/brakes.
5	Use trailer guidance and hitch-angle feedback conservatively; pull forward to correct excessive articulation.

Payload and loading

- Ford's Camper Loading Guide remains relevant even when towing: factory options, front/rear axle loading, center of gravity, payload labels, and actual scale weights govern safe loading.
- Obtain CAT-scale records with truck-only and loaded trailer configurations; retain individual axle weights and compare with door-label ratings.

11. Troubleshooting

Placeholder symptom index

Symptom	First checks	Authoritative source
Camera image missing	Power cycle; configuration history; camera connections	Owner manual / REF-008 / Ford WSM
Trailer brakes or lights inoperative	Connector, trailer wiring, fuse/relay, controller, trailer fault	Owner manual / Ford WSM
No-start or low battery	Battery condition, connections, charging system, codes	Owner manual / Ford WSM
Towing handling concern	Loading, tire pressures, hitch/trailer ratings, scale weights	Owner manual / hitch manual

12. 3D-Printed Accessories

Design library

- Preferred materials for interior vehicle parts: PETG, ASA, or ABS rather than PLA because of heat exposure.
- Candidate projects: center-console organizer/divider, under-seat modular bins, fuse kit, trailer-adapter holder, recovery-gear organizer, compressor organizer, hitch-pin rack, and bed tool clips.
- Document each design with project ID, material, printer settings, hardware, STL/CAD filename, revision notes, fitment, and installation photos.

Design record template

Field	Record
Project ID / name	
Material / color	
File and revision	
Hardware / installation	
Fitment / lessons learned	

13. Reference Documents

Source library

ID	Document	Classification / purpose
REF-001	2024 Ford Super Duty Owner's Manual, July 2023	Ford OEM; operation, maintenance, specifications, safety
REF-002	2024 Ford Camper Loading Guide	Ford OEM; payload, axle loading, center of gravity
REF-003	Window sticker	VIN-specific; retain original and extract verified build data
REF-004	Original dealer advertisement	VIN-specific archive; preserve factual metadata and photos
REF-005	Vehicle photos	VIN-specific documentation
REF-006	2024 Ford Super Duty Quick Reference Guide	Ford OEM; operating quick reference
REF-007	2023 Super Duty Fifth-Wheel Hitch Owner Manual & Installation	Ford OEM; hitch operation, parts, installation guidance
REF-008	Rear Camera Kit Enable/Disable Procedure	Ford service procedure; APIM/BCM configuration
REF-009	Spare Tire Lock document	Reference; retain with vehicle security documents
REF-010	Wheel / Tire Kit document	Reference; retain and review fitment/ratings
REF-011	2024 Super Duty Quick Start Guide	Ford OEM quick-start reference
REF-012	2024 Wrecker Towing Guide	Ford reference; recovery/towing guidance

Future source intake

Add the Ford Workshop Manual and wiring excerpts as sourced material becomes available. Record publication date, section, vehicle applicability, and any superseding publication.

Appendix A. Future Workshop Manual Intake

Required additions

- Full verified torque atlas, including sequences, torque-angle steps, and one-time-use fasteners.
- VIN-correct fluid capacities, Motorcraft product specifications, service intervals, and procedures.
- Wiring diagrams, connector views, fuse/relay functions, diagnostic tests, and module/programming references.
- OEM parts diagrams and Ford/Motorcraft part numbers.
- Service Bulletins, recalls, customer-satisfaction programs, and revision history.

Revision record

Version	Date	Summary
1.0	September 2026	Initial compiled reference; placeholders retained for WSM data